Shirsendu Chatterjee

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Education

Vellore Institute of Technology May 2023 – Present
B.Tech in Computer Science and Engineering (AI & ML) CGPA: 8.62

Technical Skills

Programming: Python, Git, Docker, GitHub Actions (CI/CD)
Machine Learning: Supervised/Unsupervised Learning, Model Evaluation, Feature Engineering, EDA, scikit-learn, Pandas, NumPy
Deep Learning: Transformers (BERT/GPT), ViT, CNNs, LSTMs, PyTorch, TensorFlow/Keras, Hugging Face, OpenCV
Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, Hybrid Retrieval (BM25 + Dense), Agentic AI, LangChain, LangGraph, SentenceTransformers
Vector Databases & Backend: FAISS, ChromaDB, FastAPI, REST APIs, PyTest
Cloud & MLOps: Google Vertex AI (Studio, Workbench, Vector Search), Google GenAI SDK, Model Deployment & Monitoring

Experience

Google Cloud Generative AI Virtual Internship — SMARTBRIDGE — Certificate Nov 2025 – Jan 2026
• Engineered end-to-end vector search pipelines on Vertex AI over a 3-month program, fusing sparse and dense embeddings (`text-embedding-005`) and deploying Vector Search indexes, endpoints, and HybridQuery retrieval flows.
• Developed multimodal RAG workflows in Vertex AI Workbench using the Google GenAI SDK: document chunking, automated extraction, grounded retrieval, and citation-backed generation over text and image inputs with Gemini.
• Applied prompt engineering and model tuning in Vertex AI Studio to optimize LLM output quality, and studied MLOps for GenAI covering deployment pipelines, continuous monitoring, and model governance.

Projects

Financial Filing RAG — Hybrid Retrieval over SEC EDGAR (GitHub) Dec 2025 – Jan 2026
• Architected a production-oriented RAG system for querying 100+ page SEC 10-K/10-Q filings with strictly decoupled ingestion and serving layers — persisted per-company indexes ensure the chat path never re-downloads or re-embeds documents.
• Implemented section-aware chunking preserving filing hierarchy (Risk Factors, MD&A, Financial Statements) with 6 metadata fields per chunk (ticker, company, filing type, date, section, index) to power precise citation-backed answers.
• Constructed hybrid retrieval fusing top-20 BM25 lexical and top-20 dense `bge-small-en-v1.5` ChromaDB results (40 deduplicated candidates), orchestrated as 5 LangGraph nodes (query understanding, retrieval, generation, citation, confidence scoring) over a provider-agnostic LLM layer (Groq, OpenAI, Gemini, Ollama).
Technologies: Python, LangGraph, LangChain, ChromaDB, BM25, SentenceTransformers, edgartools

AI Software Engineering Agent — Multi-Agent SDLC Automation (GitHub) Apr 2026 – Jun 2026
• Architected an autonomous 8-agent LangGraph state machine (Planner, Coder, Tester, Reviewer, Security, Performance, Documentation, Deployment) converting a 1-line natural-language requirement into code, tests, docs, and deployment artifacts.
• Implemented 3 feedback loops (failing tests, rejected reviews, flagged vulnerabilities) routing state to the Coder for bounded self-correction, executing generated PyTest suites in isolated workspaces with 2-layer security scans (static regex + LLM review).
• Delivered a FastAPI backend exposing 3 REST endpoints (run creation, status polling, zip download) with a zero-build-step live dashboard, a provider-agnostic OpenRouter LLM client, and Docker Compose deployment with GitHub Actions CI.
Technologies: Python, LangGraph, FastAPI, PyTest, Docker, GitHub Actions, OpenRouter

GitHub Issue Fixer — Code-Aware Agentic RAG (GitHub) Feb 2026 – Apr 2026
• Designed an AST-based semantic chunker splitting Python repositories along function/method/class boundaries (10+ file types), embedding LLM-generated chunk summaries via a code-search encoder (`st-codesearch-distilroberta-base`) in FAISS.
• Engineered an agentic multi-hop retrieval loop in LangGraph where the LLM self-evaluates evidence sufficiency and issues refined queries until adequate repository context is gathered, replacing single-shot retrieval.
• Orchestrated issue classification into 4 categories (bug, feature, documentation, security) routed to specialist prompts, with conditional Wikipedia grounding and automated Markdown responses posted via the GitHub API.
Technologies: Python, LangGraph, FAISS, SentenceTransformers, PyGithub, Groq

Achievements & Certifications

• NASA International Space Apps Challenge 2024 — Awarded “Galactic Problem Solver” certificate for Earth and space solutions.